

- 10 (a)** product of density and speed of sound / wave
 (density of medium and) speed of sound / wave in medium M1
 A1 [2]
- (b)** if $(Z_1 - Z_2)$ is small, mostly transmission M1
 if $(Z_1 - Z_2)$ is large, mostly reflection M1
(if 'mostly' not stated allow 1/2 marks for these first two marks)
 either reflection / transmission also depends on $(Z_1 + Z_2)$
 or intensity reflection coefficient = $(Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$ A1 [3]
- (c)** e.g. smaller structures can be distinguished B1
 because better resolution at shorter wavelength / higher frequency B1 [2]